A Machine Learning Approach to Minimizing Airline Losses & Improving Customer Satisfaction

Phase 3

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Outline

- Abstract
- Project Description
- Summary Visual EDA
- Feature Engineering
- Modeling Pipelines
  - Results
  - Discussion
- Conclusion

Abstract

- **Problem Impact**
  - Flight delays cost airlines billions annually and disrupt passenger travel
- **Initial Scope**
  - Modeling focused on 3+ hour delays per DOT refund policy
- **Data**
  - Models were trained on enriched datasets combining flight, weather, and airport data.
- **Baseline**
  - Logistic Regression served as a strong baseline for model performance
- **Optimization**
  - Random Forest was tuned using Optuna
- **Neural Network Performance**
  - Multilayer Perceptron models were consistent but underperformed relative to other models
- **Recommendation**
  - XGBoost

Project Description

Motivation

- Flight delays cost billions annually and disrupt operations, infrastructure, and passenger experience.
- Proactive delay prediction helps optimize airline logistics, improve airport coordination, and inform passengers in advance.

Audience

- Airlines
- Airport Authorities
- Air Traffic Controllers
- Passengers & Travel Tech

Summary Visual EDA

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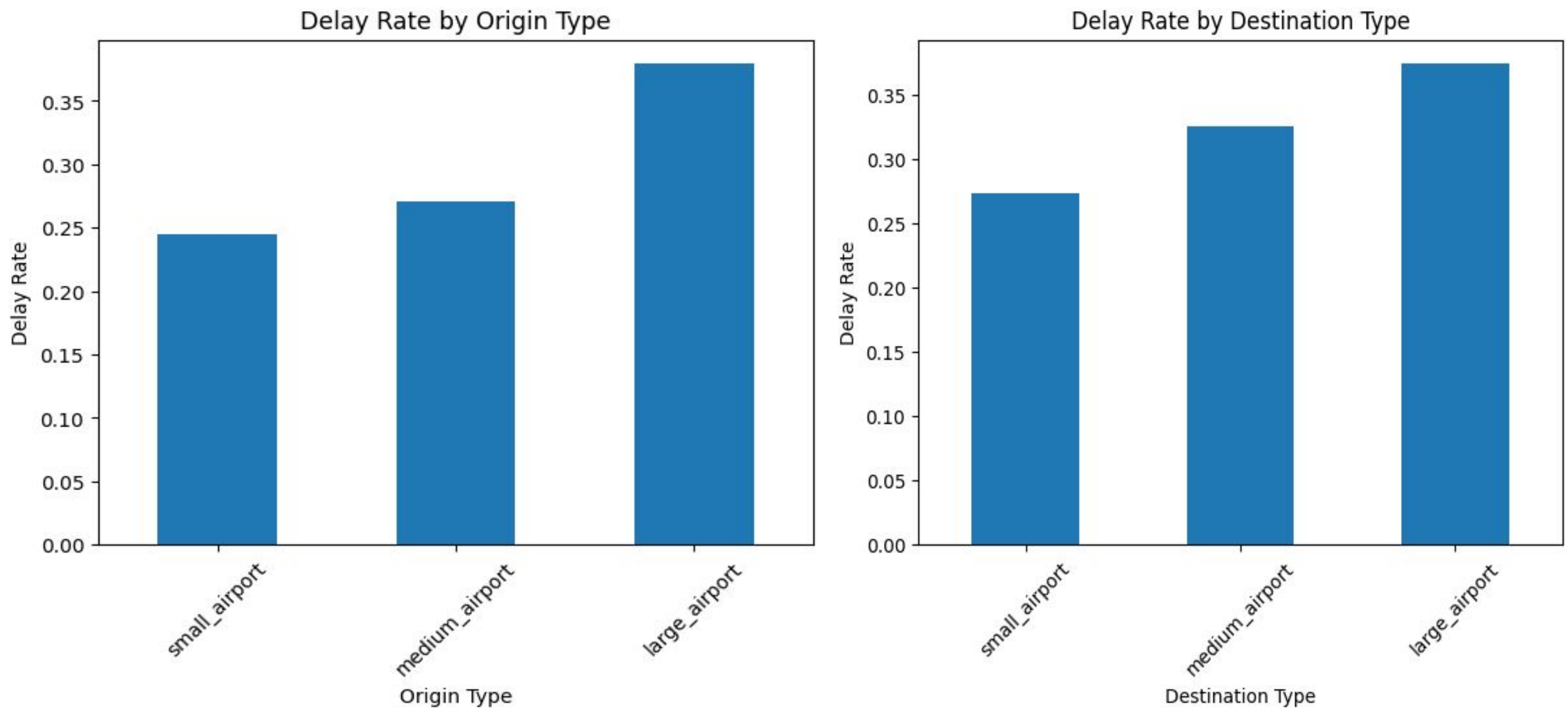
Summary Visual EDA



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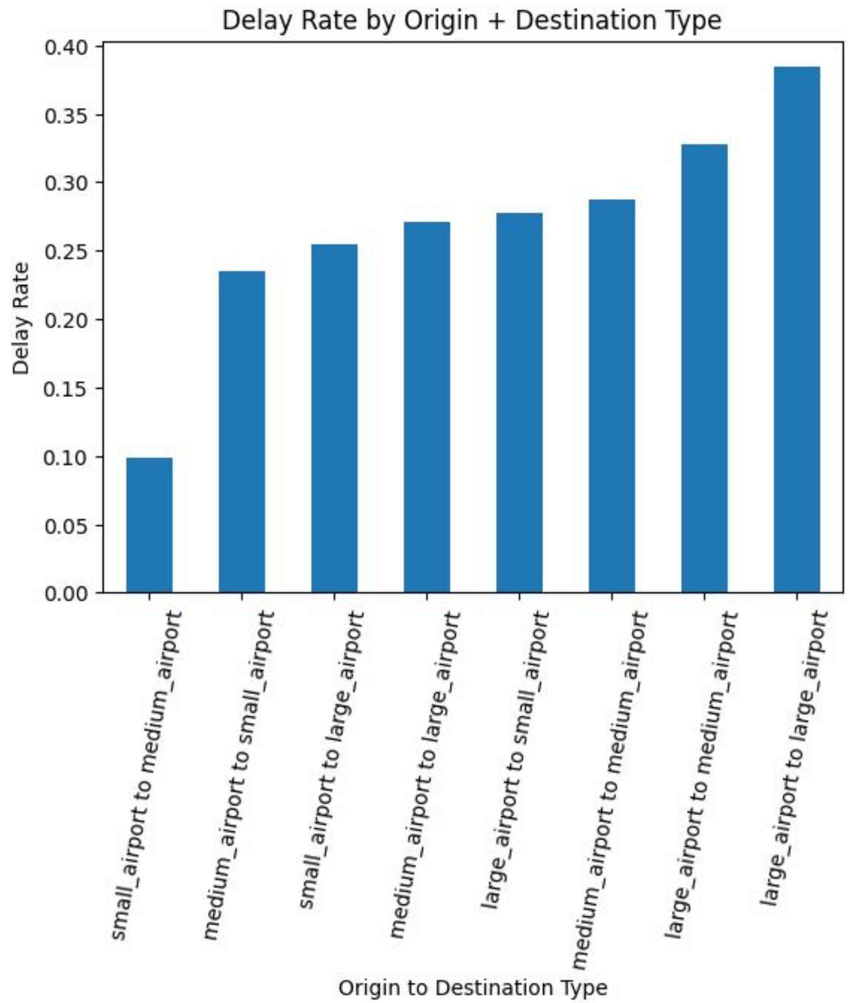
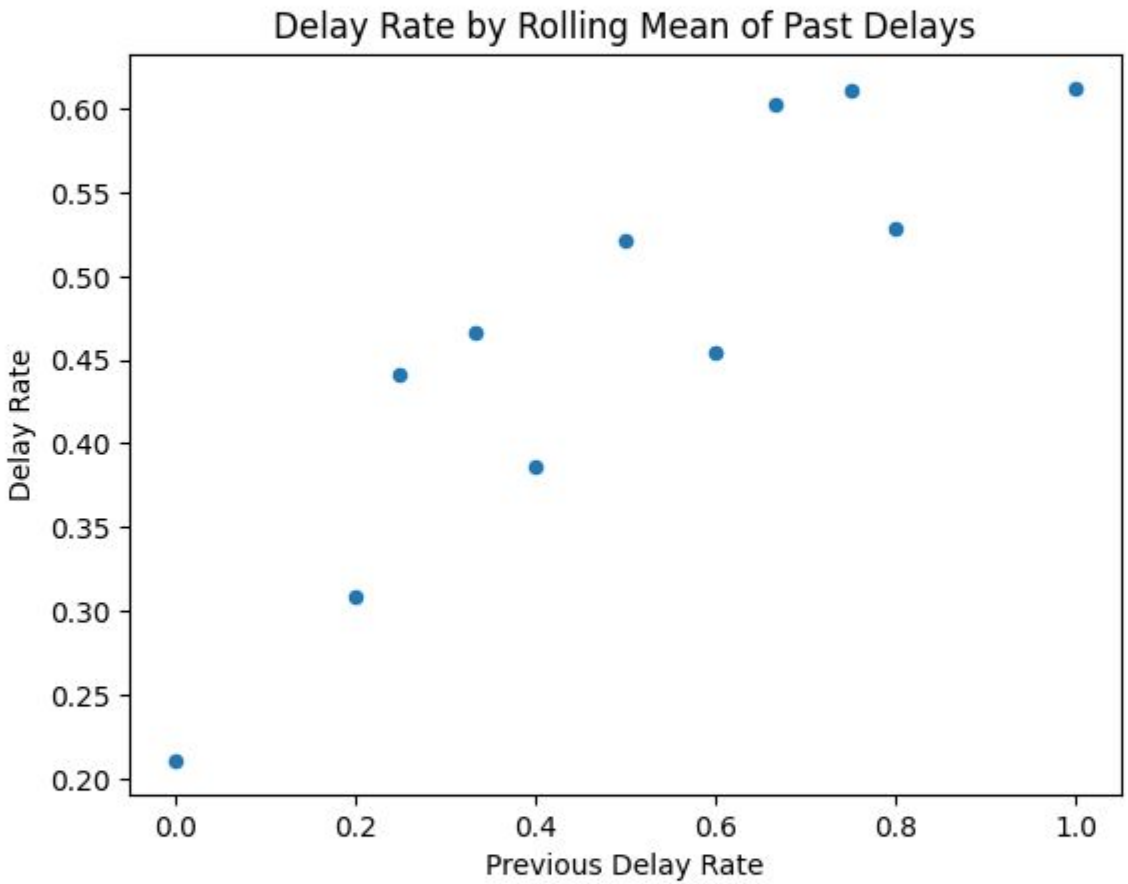

Delay Rate by Origin + Destination Type
Delay Rate
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
small_airport to medium_airport
medium_airport to small_airport
small_airport to large_airport
medium_airport to large_airport
large_airport to small_airport
medium_airport to medium_airport
large_airport to medium_airport
large_airport to large_airport
Origin to Destination Type

Summary Visual EDA



$$\bar{y}_t = \frac{1}{T}\sum_{\tau=1}^{T} y_{t-\tau}$$

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Feature Engineering

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Features

- Drop features based on correlation and ANOVA test
- Imputations
  - ➢ group-based average
  - ➢ median
  - ➢ weather columns imputed by taking the average of a 1 hour window
- Derived features
  - ➢ hour of schedule departure
  - ➢ is weekend
  - ➢ is holiday
  - ➢ is rush hour
  - ➢ delay rate by rolling mean of past delays
  - ➢ delay rate by origin + destination type
  - ➢ is rainy
  - ➢ log distance
  - ➢ log elevation
- Train (2015-2018) / Test(2019) splits

Time Series Cross Validation - Optuna

- **Models Tuned:** Random Forest, XGBoost, Multi-Layer Perceptron (MLP)
- **Validation Strategy:** 3-fold time-based CV with growing training window, fixed test size (30K)
- **Imbalanced Data Handling:** Undersampling majority class to 4:1 ratio
- **Metric Optimized:** F1 Score
- **Tuning Framework:** Optuna with TPE Sampler
  - Tree-structured Parzen Estimator
- **Best Model Selection:** Based on highest average F1 from Optuna trials

Best Performing Parameters

- **Random Forest (F1=0.4831)**
  - num_trees: 15
  - max_depth: 18
- **XGBoost (F1=0.4876)**
  - eta: 0.113
  - max_depth: 10
  - min_child_weight: 3
  - subsample: 0.599
  - colsample_bytree: 0.808
  - gamma: 0.388
- **Multi-Layer Perceptron**
  - Didn't have results due to various reasons
  - We have CV framework set up for MLP, not enough time to complete
    - it's taking more than 20 hours to finish a trial
  - Continue to run and reflect the result in final report

Modeling Pipelines Explored

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Baseline - Logistic Regression

- ## Model Comparison

Model	Wall Time (Seconds)	Number of Features	Regularization	Features
Model LR-G	320.48	33	Lasso	scaled_weather_features (12) + 'Day_OF_MONTH', 'CRS_DEP_TIME', 'TAXI_IN', 'TAXI_OUT', 'WHEELS_ON', 'WHEELS_OFF', 'DIVERTED', 'CRS_ARR_TIME', 'CRS_ELAPSED_TIME' + 6 derived features (`'log_distance'`, `'log_elevation'`, `'IS_WEEKEND'`, `'IS_PEAK_HOUR'`, `'is_holiday'`, `'CRS_DEP_HOUR'`) + encoded categorical variables ('OP_UNIQUE_CARRIER', 'ORIGIN', 'origin_type', 'dest_type', 'DEST', 'DEP_TIME_BLK')
Model LR-H	431.87	33	Ridge	scaled_weather_features (12) + 'Day_OF_MONTH', 'CRS_DEP_TIME', 'TAXI_IN', 'TAXI_OUT', 'WHEELS_ON', 'WHEELS_OFF', 'DIVERTED', 'CRS_ARR_TIME', 'CRS_ELAPSED_TIME' + 6 derived features (`'log_distance'`, `'log_elevation'`, `'IS_WEEKEND'`, `'IS_PEAK_HOUR'`, `'is_holiday'`, `'CRS_DEP_HOUR'`) + encoded categorical variables ('OP_UNIQUE_CARRIER', 'ORIGIN', 'origin_type', 'dest_type', 'DEST', 'DEP_TIME_BLK')
Model LR-I	314.38	34	Mix of Lasso and Ridge	scaled_weather_features (12) + 'Day_OF_MONTH', 'CRS_DEP_TIME', 'TAXI_IN', 'TAXI_OUT', 'WHEELS_ON', 'WHEELS_OFF', 'DIVERTED', 'CRS_ARR_TIME', 'CRS_ELAPSED_TIME' + 7 derived features (`'log_distance'`, `'log_elevation'`, `'IS_WEEKEND'`, `'IS_PEAK_HOUR'`, `'is_holiday'`, `'CRS_DEP_HOUR'`, `'is_rainy'`) + encoded categorical variables ('OP_UNIQUE_CARRIER', 'ORIGIN', 'origin_type', 'dest_type', 'DEST', 'DEP_TIME_BLK')
Model LR-J	1609.83	33	None	scaled_weather_features (12) + 'Day_OF_MONTH', 'CRS_DEP_TIME', 'TAXI_IN', 'TAXI_OUT', 'WHEELS_ON', 'WHEELS_OFF', 'DIVERTED', 'CRS_ARR_TIME', 'CRS_ELAPSED_TIME' + 8 derived features (`'log_distance'`, `'log_elevation'`, `'IS_WEEKEND'`, `'IS_PEAK_HOUR'`, `'is_holiday'`, `'CRS_DEP_HOUR'`, `'rolling_delay_mean'`) + encoded categorical variables ('OP_UNIQUE_CARRIER', 'ORIGIN', `'ORIGIN_DEST_TYPE'`, 'DEST', 'DEP_TIME_BLK')

- ## Model Performance Metrics

Model	AUC	AUPR	Precision	Recall	F1 Score
Model LR-G	0.6416	0.4739	0.4443	0.5993	0.5103
Model LR-H	0.6906	0.5415	0.4888	0.6324	0.5514
Model LR-I	0.6420	0.4742	0.4445	0.5992	0.5103
`Model LR-J`	`0.7179`	`0.5858`	`0.5165`	`0.6442`	`0.5733`

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Random Forest

- Model Performance Metrics

Model	Wall Time (Seconds)	Max Depth	Num Trees	Max Bins	Min Instances Per Node	AUC	AUPR	Precision	Recall	F1-Score
Model RF-G	1495.63	15	50	128	Default	0.7278	0.6119	0.4774	0.7233	0.5752
Model RF-H	1140.98	15	50	Default	Default	0.7266	0.6046	0.4782	0.7265	0.5768
Model RF-I	2994.91	15	100	128	5	0.7271	0.6049	0.4819	0.7215	0.5779

- Model Performance Metrics After Fine Tuning with OPTUNA

Model	Wall Time (Seconds)	Max Depth	Num Trees	Max Bins	Min Instances Per Node	AUC	AUPR	Precision	Recall	F1-Score
Model RF-J	**742.28**	**18**	**15**	**Default**	**Default**	**0.7502**	**0.6317**	**0.5246**	**0.7055**	**0.6017**
`Model RF-J Version 2`	`2989.04`	`25`	`15`	`Default`	`Default`	`0.7759`	`0.6742`	`0.5640`	`0.6973`	`0.6236`

XGBoost

- Results

Model	Max Number of Trees	Wall Time (seconds)	AUC	AUPR	Precision	Recall	F1
XGBoost	1000	6300	0.9598	0.9382	0.6682	0.9597	0.7869

- Early Stopping Hyperparameters:
  - Patience: 10 boosting rounds
  - Evaluation Metric: Cross Entropy Loss
  - Training Set: First 3 Years
  - Validation Set: Year 4

Multi-layer Perceptron

- Results

	Model	Recall	Precision	F1 Score
2 Hidden Layers	**A**	0.657	0.412	0.521
	B	0.599	0.359	0.449
	C	0.657	0.432	0.521
1 Hidden Layer	**A**	0.657	0.432	0.521
	B	0.599	0.357	0.449
	C	0.657	0.412	0.521

- Fine Tune Baseline Pipeline with Grid Search
  - Fine Tuning with grid search is still running in the cluster
  - Some manual investigation found the following to be the best parameters:

Model	num_neurons_layer1	num_neurons_layer2	max_itr	step_size	AUC	AUPR	Precision	Recall	F1
C - Version 5	64	48	100	0.03	0.3695	0.5923	0.4086	0.2812	0.3345

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Conclusions

- **XGBoost** achieved the highest F1 score of **78.69%**, showing strong predictive performance on an imbalanced real-world dataset.
- Model can be deployed in airline operations to flag potential delays early and support proactive decision-making.
- Limitations include:
  - External factors like air traffic control/staffing not captured.
  - Requires real-time access to consistent, region-wide data.
- Opportunities for improvement:
  - Enhanced feature engineering
  - Additional data sources (e.g., air traffic)
  - Further hyperparameter fine-tuning

Questions?